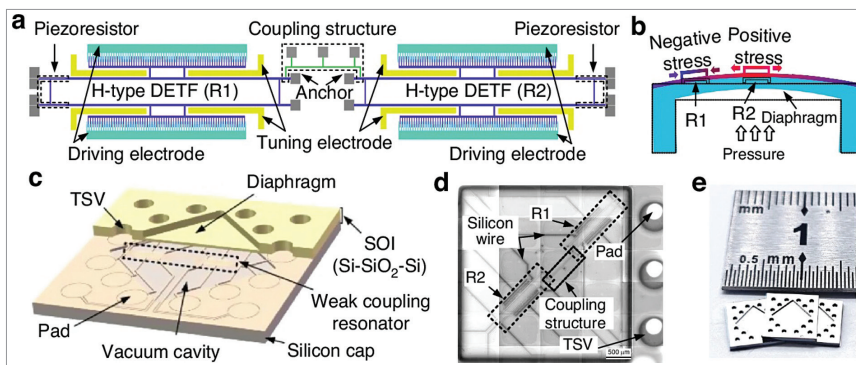


Tiny dual-mode MEMS vacuum sensor

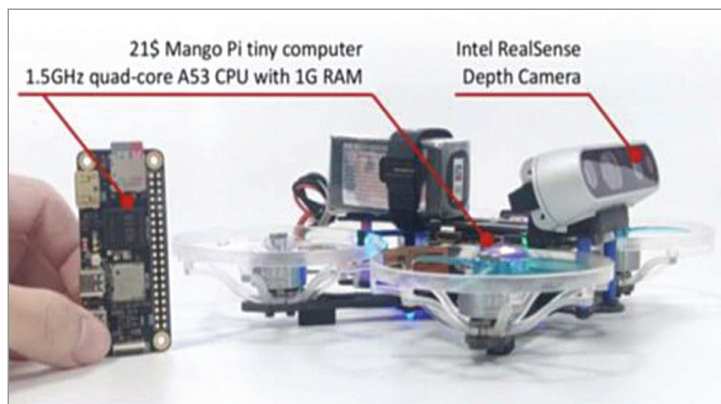


(a) Structure of the proposed weak-coupling resonator, (b) Schematic of diaphragm deformation and stress under the loaded pressure, (c) Structure of the sensor chip, (d) Infrared microscope image of the fabricated sensor from diaphragm side, (e) The picture of the fabricated sensor chip (Credit: Microsystems & Nanoengineering, 2025)

A research team at Aerospace Information Research Institute (AIR) of the Chinese Academy of Sciences (CAS), (Beijing, China) has developed a grain-sized dual-mode MEMS vacuum sensor that delivers high precision and ultra-wide pressure detection. It switches intelligently between two resonator modes, offering 0.1Pa resolution and extreme thermal stability. Unlike bulky hybrids, it is gas-agnostic and over 200x smaller, ideal for semiconductor fabs and aerospace systems demanding compact, accurate, and reliable vacuum sensing.

Compact swarm-level AI drones navigation using neural network

Researchers at Shanghai Jiao Tong University (Shanghai, China) have developed a compact AI navigation system for drones. It uses a lightweight neural network trained with physics, not maps or labelled data, enabling drones to fly autonomously using just a 12×16 depth map. Running on a \$21 board, it supports high-speed, swarm-level navigation without communication. The result: scalable, stable drone autonomy for complex missions, even in GPS-denied, sensor-light environments like rescue ops and inspections.



The team's neural network is run on a low-end development board that costs only 21\$ (Credit: Zhang et al)

Public USB charging ports can now hack phones

A new threat called choicejacking bypasses 'charge only' prompts at public USB ports by impersonating input devices and injecting commands in just 133 milliseconds. Developed by researchers at Graz University (Styria, Austria), this exploit forces phones into data-transfer

mode, risking data theft and malware infection without user consent. Unlike juice jacking, it targets both iOS and Android at a deeper level. The safest move? Avoid public USBs, use personal chargers, power banks, and keep devices updated.